

ANIL PRADHAN

BACKEND ENGINEER | LLM APPLICATION DEVELOPER

Backend-focused Computer Science undergraduate with hands-on experience building production-grade AI applications using FastAPI, LangGraph, RAG, Docker, and cloud technologies. Passionate about scalable backend architecture, distributed systems, and AI-powered software.

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Bhubaneswar, Odisha

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)



EDUCATION

Institute of Technical Education & Research (ITER)

Siksha 'O' Anusandhan (SOA) University, Bhubaneswar

Bachelor of Technology (B.Tech), Computer Science & Engineering

2024 - 2028 | CGPA: 9.11/10

Council of Higher Secondary Education (CHSE), Odisha

Class XII (PCM with IT)

2022 - 2024 | 85%



EXPERIENCE

College Ambassador (Corporate Projects Program) | Techfest, IIT Bombay

Jul 2026 - Present

- Selected as a College Ambassador for Techfest, IIT Bombay, representing one of Asia's largest science and technology festivals.
- Completed industry-sponsored corporate projects across web development, business strategy, and market research in collaboration with partner organizations.
- Qualified for Round 2 of the FlowFinance Growth Strategy & Business Development Challenge by developing a business strategy presentation and market analysis.
- Collaborated with multidisciplinary teams while delivering project submissions within defined deadlines.



TECHNICAL PROJECTS

AI Career Mentor | Live Demo | Solo Full-Stack Developer

March 2026 - Present

- Multi-Agent Orchestration & Concurrency:** Orchestrated 7 specialized AI agents via a parallel LangGraph DAG, designing a two-phase concurrency flow (concurrent resume parsing and market scraping, followed by parallel LinkedIn and career roadmap generation) that reduced end-to-end latency by 60% (4 min → 60s).
- Real-Time Bidirectional Streaming & FSMs:** Engineered a full-duplex WebSocket communication pipeline utilizing the Gemini Multimodal Live API (gemini-2.5-flash-native-audio-latest) for real-time voice coaching, governed by a 7-phase Finite State Machine (FSM) that dynamically manages mock interview stages and Monaco IDE code evaluations.
- API Resilience & Failover Architecture:** Built a resilient multi-provider LLM fallback chain (Groq ⇄ Cerebras ⇄ NVIDIA NIM) with custom circuit breakers and daily rate-limiting/gap-locking policies, maintaining 100% service availability and zero request drops during peak API rate-limit spikes.
- Resource-Optimized RAG & Quality Control:** Optimized RAG retrieval by deploying local ONNX embeddings (all-MiniLM-L6-v2) to eliminate API dependency, implementing a keyword-search fallback to prevent OOM crashes on memory-constrained (512MB RAM) servers, backed by a 116-test Pytest CI/CD suite.

FastAPI

LangGraph

ChromaDB

ONNX

Docker

Upstash Redis

Next.js 14

Gemini Live

Task Pilot-AI | Live Demo | Integration Lead • Team IdeaForg-E

June 2026 - July 2026

- Heterogeneous Data Aggregation:** Architected an ingestion pipeline that normalizes and aggregates raw developer data from 7 fragmented sources (Jira, GitHub, Slack, Email, Calendar, Meetings, and Incidents) into a unified relational database, saving developers ~2 hours daily of manual task tracking.
- Hybrid Extraction & Deduplication Engine:** Designed a cost-optimized extraction and fusion pipeline using deterministic heuristic models, fuzzy text correlation (SequenceMatcher), and LLM reasoning (Qwen-3.6-27B) to extract hidden action items from unstructured sources, recovering 35% of untracked tasks while reducing LLM API token costs by 75%+.
- Concurrency-Safe Database & High-Priority API:** Configured SQLite in Write-Ahead Logging (WAL) mode with custom busy-timeout strategies, facilitating concurrent multi-agent writes with zero database locking, integrated with a real-time P1 incident-injection API that autonomously re-prioritizes the 6-stage pipeline in under 15 seconds.
- Self-Healing Orchestration & Quality Auditor:** Developed a background daemon to monitor execution states (WorkflowRun) and automatically self-heal stale runs (stuck runs >5 mins), combined with a 7-dimension automated Quality Agent that grades tasks on actionability and generates context-aware follow-up queries for missing information.

FastAPI

LangGraph

SQLite

LLMs

React

Redis



TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL

Backend: FastAPI, LangGraph, Pydantic, SQLAlchemy 2.0, WebSocket, Prometheus, Sentry, pdfplumber, edge-tts

Databases: Neon Postgres, SQLite, Upstash Redis, ChromaDB (Vector Store)

DevOps & Infra: Docker, GitHub Actions (CI/CD), Vercel, Render.com

Frontend: HTML, CSS, Tailwind CSS, Axios, React

AI / LLM & RAG: Groq Cloud, Cerebras Cloud, NVIDIA NIM, Gemini Live API, Tavily, Serper.dev, DuckDuckGo

Auth & Security: JWT (python-jose), Google OAuth 2.0, bcrypt, CORS Middleware, Rate Limiting, HTTPS

Tools: VS Code, Git, GitHub, Swagger UI, Postman



ACHIEVEMENTS & CERTIFICATIONS

- Google Cloud Arcade Program (2024-25): Completed GCP challenges, earning 75 Arcade Points across hands-on cloud labs.
- Solved 200+ problems across GFG and LeetCode, awarded GFG Certificate and T-shirt.
- Certifications: Credentials in Python, AI, and Web Dev from IBM, Cisco, Anthropic, Infosys, Udemy.
- Competitions: Participated in multiple IIT/NIT coding contests and hackathons focused on AI systems, backend engineering, and algorithmic problem solving.
- Hackathons: Built production-grade MVP solutions for Microsoft AI DevDays (\$80K+), Amazon Nova AI (\$95K+), and Google Solution Challenge 2026.
- Active participant in Developer Communities (GDG Events) and college tech events, exploring AI systems, cloud infrastructure, and backend engineering.